

Which limits of Newton's construction reach the zero branch

The thesis under test (user, 2026-10-01): Newton's mechanics exists only with a positive action scale, no zero branch is continuously reachable, and the family of theories has a gap in the sense of the mass gap. This note states the thesis as a claim about limits and proves, in the simplest cases, which of Newton's own limits reach the zero theory and which do not. Result: with Gaussian preparations and Gaussian marks, Galileo's comparison has a record family that is the same for every value of the constant (Theorem 1), so there is no gap there. For a determinate preparation, one definite velocity at each place, the complete coordinate record has no limit as the constant tends to zero once the preparation has folded, or a limit different from Newton's, while every smeared record converges to Newton's branch-weighted density (Theorem 2, any finite partition; Theorem 3, exactly, for inertial motion with a graded velocity after its fold time). The gap thesis therefore holds in the topology of complete records of determinate preparations, beyond their first fold, and fails in the topology of smeared records or for statistical preparations. Refinement with records is the one Newtonian limit in which the zero theory is the regular side. Proofs are written derivations by Claude Fable (2026-10-01), unrefereed; the stationary phase and almost-periodicity facts used are standard and named.

Throughout, h denotes the reduced phase constant, as in the 1998-conjecture note; the recording notes write $\kappa = h/2$.

1. The thesis as a statement about limits

For each $h > 0$ let T_h be the theory with phase constant h : its dynamics and its record law. The dilation $h \mapsto \mu h$ is an isomorphism (fifth-postulate note, Theorem B and §7; Rivero 1998, eq. (10), full read), so $\{T_h : h > 0\}$ is one theory in different units. The zero theory T_0 is Newton's. The thesis says that T_0 is not the limit of T_h as $h \rightarrow 0$ in any topology intrinsic to Newton's construction: the scale is generated by the construction, and zero is isolated. This is the mechanical form of a mass gap, where the continuum limit generates the scale and the massless theory is not reached by the coupling.

Known results fix what the thesis can mean. The limit $h \rightarrow 0$ is singular, with phases $e^{iS/h}$ that have no pointwise limit [Berry2002; metadata]. Yet in weak topologies and for finite times the classical theory is reached: coherent states follow classical trajectories for finite times [Hepp1974; metadata], Egorov's theorem transports smooth observables classically up to $O(h)$ for finite times, and Wigner measures give the weak limits of densities [LionsPaul1993; metadata]. So the thesis is false for smeared records at finite time, and can be true only in stronger topologies. Newton's construction supplies three candidates: complete records, the densities themselves rather than their smearings; the all-time limit; and refinement, with or without records at the inserted times. The rest of the note tests each.

Deformation families cannot see a gap. The Moyal family, the Fisher field law and its sheet realizations (score-constrained note, §§6–8) are continuous in the constant by construction and contain the zero branch. The tangent groupoid glues the $\varepsilon = 0$ fibre continuously to the $\varepsilon > 0$ fibres (fifth-postulate note, §7), so the kinematics is continuous at zero as well. A gap, if it exists, lives in the dynamics and the records.

2. Theorem 1: Galileo's comparison has no gap

Theorem 1 (Gaussian preparations, quadratic forces, Gaussian marks). Let $H = p^2/2m + V$ with $\deg V \leq 2$ (inertia, uniform gravity, Hooke's force). For each admissible $h \geq 0$ prepare the body with the Gaussian phase-space law $N(z_0, \Sigma)$, $\Sigma > 0$ fixed; for $h > 0$ this is the Gaussian state with that Wigner function, which exists iff $\det \Sigma \geq h^2/4$. Mark the body at times t_j by Gaussian coordinate copies of widths σ_j , each read or forgotten. Then the joint law of all read records and of the final coordinate is, for every admissible h , the classical Gaussian process

of T_0 with one addition: each mark, read or not, adds to the body momentum an independent centred Gaussian kick of variance $h^2/(4\sigma_j^2)$. The family is continuous in h and equals T_0 at $h = 0$.

Proof. For $\deg V \leq 2$ every Moyal flow is the classical affine symplectic flow, and the Wigner function is transported classically (fifth-postulate note, Theorem A(b)). A coordinate copy is generated by $q\pi$, quadratic in the joint canonical variables, so the same holds for the body–pointer shear. A Gaussian Wigner function of a product state is a positive density, and reading the pointer coordinate is classical conditioning of that density. The pointer’s momentum π has variance $h^2/(4\sigma_j^2)$ and is independent of its coordinate for a minimal Gaussian pointer, so the shear transfers to the body a kick of that variance whether or not the coordinate is read; this is Proposition 1 of the parallel-move note for the forgotten case, and the read case adds the classical Gaussian conditioning on $r = q + z$. Composition in time is the classical flow between marks. The variance $h^2/(4\sigma_j^2)$ is continuous in h and zero at $h = 0$. $\square$

So on the comparison as the Planck paper sets it, the record family is constant in h up to continuous kicks. The zero branch is reached in every topology: there is no gap in Galileo’s parabola with statistical preparations. The same holds for refinement with records at every inserted time: the fixed- h limit exists iff $\sum_j \sigma_j^{-2} < \infty$ (parallel-move note), the $h = 0$ limit always exists, and the $h \rightarrow 0$ limit at any fixed partition is continuous. In this limit the zero theory is the regular side and $h > 0$ the singular one.

3. Theorem 2: determinate preparations with several paths

Take the symmetric discrete action of the 1998-conjecture note, (1) there, on a partition with N cells of lengths τ_j , smooth V , fixed initial point $q_0 = x$, record variable $y = q_N$, and internal vertices $z = (q_1, \dots, q_{N-1}) \in \mathbb{R}^d$, $d = N - 1$. With a cutoff $\chi \in C_c^\infty(\mathbb{R}^d)$ and a smooth amplitude O , the halved functional with the record variable kept free is

$$A_h(y) = (2\pi h)^{-d/2} \int \chi(z) O(z) e^{iS_\pi(x, z, y)/h} dz. \quad (1)$$

Its modulus squared $|A_h(y)|^2$ is the complete coordinate record at the final time; $\int \varphi(y) |A_h(y)|^2 dy$ with $\varphi \in C_c^\infty$ is a smeared record.

Hypothesis (several nondegenerate paths). On an open interval Y of record values, $S_\pi(x, \cdot, y)$ has exactly $K \geq 2$ critical points $z_a(y)$ in $\text{supp } \chi$, all nondegenerate, with $\chi = 1$ near each. By the implicit function theorem they depend smoothly on y . Write

$$S_a(y) = S_\pi(x, z_a(y), y), \quad c_a(y) = \frac{O(z_a(y)) e^{i\pi\sigma_a/4}}{|\det S''_\pi(z_a(y))|^{1/2}}, \quad (2)$$

with σ_a the Hessian signature.

Lemma (final momenta are distinct). $p_a(y) := S'_a(y)$ equals $m(y - q_{N-1}^{(a)}(y))/\tau_{N-1} - \frac{1}{2}\tau_{N-1}V'(y)$, and $p_a(y) \neq p_b(y)$ for $a \neq b$.

Proof. The envelope formula gives $S'_a(y) = \partial_y S_\pi$ at the critical point, which is the displayed expression. The discrete Euler–Lagrange equation at vertex j , $m(q_{j+1} - q_j)/\tau_j - m(q_j - q_{j-1})/\tau_{j-1} + \frac{1}{2}(\tau_{j-1} + \tau_j)V'(q_j) = 0$, is linear in q_{j-1} with coefficient $m/\tau_{j-1} \neq 0$. So a critical point is determined by (q_{N-1}, q_N) by backward recursion, two distinct critical points have distinct q_{N-1} , and the momenta differ. $\square$

Theorem 2. Under the hypothesis, uniformly for y in compact subsets of Y ,

$$|A_h(y)|^2 = \sum_a |c_a(y)|^2 + \sum_{a < b} 2|c_a(y)c_b(y)| \cos\left(\frac{S_a(y) - S_b(y)}{h} + \theta_{ab}(y)\right) + O(h). \quad (3)$$

(i) *Complete record.* Fix $y \in Y$. The limit of $|A_h(y)|^2$ as $h \downarrow 0$ exists only if, for every nonzero value ω of $S_a(y) - S_b(y)$, the amplitudes $|c_a c_b| e^{i\theta_{ab}}$ of the pairs with that action difference sum to

zero. For $K = 2$ with $S_1(y) \neq S_2(y)$ this means $c_1(y)c_2(y) = 0$. When the limit exists it equals $\sum_a |c_a|^2$ plus the cross terms of pairs with equal actions, which is Newton's branch-weighted density $D_{0,\pi}(|O|^2) = \sum_a |O(z_a)|^2 / |\det S''_\pi(z_a)|$ only if those cross terms vanish too. (ii) *Smeared record*. For every $\varphi \in C_c^\infty(Y)$,

$$\int \varphi(y) |A_h(y)|^2 dy \longrightarrow \sum_a \int \varphi(y) \frac{|O(z_a(y))|^2}{|\det S''_\pi(z_a(y))|} dy \quad (h \downarrow 0). \quad (4)$$

Proof. Equation (3) is the stationary phase expansion of (1) with nondegenerate critical points, (3) of the 1998-conjecture note, whose remainder is uniform in the parameter y on compact sets where the critical points stay nondegenerate (Hörmander's stationary phase theorem with parameters; standard). Expanding the modulus squared gives (3) with $\theta_{ab} = \pi(\sigma_a - \sigma_b)/4 + \arg O_a - \arg O_b$.

- (i) Put $s = 1/h$. The cross sum in (3) is an almost periodic function $f(s)$ in Bohr's sense, a finite trigonometric sum. If $|A_h(y)|^2$ converges as $h \downarrow 0$ then $f(s)$ converges as $s \rightarrow \infty$. An almost periodic function with a limit at infinity is constant: for any s_0 and $\delta > 0$ the set of δ -almost periods is relatively dense, so there are $T_n \rightarrow \infty$ with $|f(s_0 + T_n) - f(s_0)| < \delta$, and the limit forces $f(s_0) = \lim f$. A constant trigonometric sum has zero coefficient at every nonzero frequency: the Bohr mean of $f(s)e^{-i\omega s}$ is that coefficient. This proves the condition, and the value of the limit when it exists.
- (ii) In the cross term with indices $a \neq b$ the phase $\psi_{ab} = S_a - S_b$ has $\psi'_{ab} = p_a - p_b \neq 0$ on Y by the Lemma. One integration by parts, $\int g e^{i\psi/h} dy = ih \int (g/\psi')' e^{i\psi/h} dy$ for $g \in C_c^\infty(Y)$, bounds the term by $O(h)$. The diagonal terms and the uniform remainder give (4). $\square$

Corollary (finite resolution). Let $\varphi \geq 0$ be smooth with compact support and $\int \varphi = 1$, and let $\varphi_\delta(y) = \delta^{-1} \varphi((y - y_0)/\delta)$ be a record of resolution δ at $y_0 \in Y$. With $\Delta p > 0$ the minimum of $|p_a - p_b|$ over pairs $a \neq b$ on the support,

$$\int \varphi_\delta |A_h|^2 dy = \sum_a \int \varphi_\delta \frac{|O(z_a)|^2}{|\det S''_\pi(z_a)|} dy + O\left(\frac{h}{\delta \Delta p}\right) + O(h). \quad (4')$$

A record of resolution δ reaches Newton's density iff $h/(\delta \Delta p) \rightarrow 0$: the fringes, of spacing $h/\Delta p$, must be unresolved. The complete-record topology of (i) is the order of limits in which $\delta \rightarrow 0$ before $h \rightarrow 0$; the smeared topology of (ii) is the opposite order.

Proof. In the integration by parts of (ii), the derivative of φ_δ carries a factor δ^{-1} relative to φ_δ itself, and $\int \varphi_\delta = 1$; the remaining terms are as before. $\square$

Thus the complete record of a determinate preparation with several paths does not reach Newton's density as $h \downarrow 0$, by non-convergence in the generic case of distinct actions and by a wrong limit in the symmetric case of equal actions, while its smearing always does. Theorem 1 of the 1998-conjecture note is the case $d = 1$, $K = 3$, $y = 0$ of (i), including the retained cross term of the two mirror paths; smearing in y removes that term by (ii), because the mirror paths arrive at $y = 0$ with opposite momenta.

4. Theorem 3: inertial motion with a graded velocity

Newton's simplest motion already shows the gap when the preparation is determinate and its velocity varies along it. Let $h > 0$, $V = 0$, and prepare

$$\psi_0(q) = a(q) e^{iS_0(q)/h}, \quad a \in C_c^\infty(\mathbb{R}) \text{ real}, \quad p_0 = S'_0. \quad (5)$$

Each particle at q has the definite velocity $p_0(q)/m$. The free propagator is exact, so

$$\psi_t(x) = \left(\frac{m}{2\pi i \hbar t}\right)^{1/2} \int a(q) e^{i\Phi_x(q)/h} dq, \quad \Phi_x(q) = \frac{m(x - q)^2}{2t} + S_0(q). \quad (6)$$

Critical points of Φ_x are the arrivals $q + p_0(q)t/m = x$, and $\Phi_x''(q) = m/t + p_0'(q)$. Define the fold time

$$t_* = \frac{m}{\max(-p_0')} \in (0, \infty], \quad (7)$$

with $t_* = \infty$ when $p_0' \geq 0$ everywhere.

Theorem 3. (a) For $0 < t < t_*$ the arrival map $q \mapsto q + p_0(q)t/m$ is a diffeomorphism, every x has one arrival $q_*(x)$, and $|\psi_t(x)|^2 \rightarrow a(q_*(x))^2/(1 + p_0'(q_*(x))t/m)$ pointwise, the classical transported density: the complete record reaches the zero branch before the fold. (b) For $t > t_*$ there is an open interval of x with three nondegenerate arrivals $q_1 < q_2 < q_3$. Where a is nonzero at two or more of them, Theorem 2 applies with $d = 1$, $K = 3$: $|\psi_t(x)|^2$ has no limit as $h \downarrow 0$ unless the cross amplitudes cancel, and every smearing in x converges to the classical three-branch density $\sum_a a(q_a)^2/|1 + p_0'(q_a)t/m|$. (c) At a fold point $x_c(t)$, where $\Phi_x''(q_c) = 0$ and $\Phi_x'''(q_c) \neq 0$, $|\psi_t(x_c)|^2$ grows like $h^{-1/3}$: the record diverges with the classical caustic density.

Proof. (a) The derivative $1 + p_0'(q)t/m$ is positive for $t < t_*$, so the arrival map is strictly increasing and onto, and the single critical point is nondegenerate; one-dimensional stationary phase in (6) gives $|\psi_t|^2 \rightarrow a^2/|\Phi_x''| \cdot (m/t)$, which is the stated ratio. (b) For $t > t_*$ the derivative is negative on an open set, so the arrival map is not monotone; by the intermediate value theorem the preimage of each x in an open interval has three points, nondegenerate away from the two fold values. The amplitudes $c_a = a(q_a)e^{i\pi\sigma_a/4}/|\Phi_x''(q_a)|^{1/2}$ and the actions $\Phi_x(q_a)$ play the roles of (2); the final momenta $\partial_x \Phi_x(q_a) = m(x - q_a)/t = p_0(q_a)$ are distinct because the q_a are distinct and $x = q_a + p_0(q_a)t/m$. Theorem 2 applies. (c) At a degenerate critical point of order three the stationary phase integral scales as $h^{1/3}$ against the prefactor $h^{-1/2}$, giving $|\psi|^2 \sim h^{-1/3}$ (Airy scaling; standard). $\square$

Newton's inertial motion, with a preparation in which faster particles start behind slower ones, folds at t_* . Before the fold the zero branch is reached by complete records; after it, never. The fringe spacing in x is $h/|p_a - p_b|$, which tends to zero without the record ever settling. The classical multi-stream density is recovered only by smearing, an operation external to the determinate construction.

5. All-time invariants

Two quantities already in the corpus show the other non-commuting pair of limits, $h \rightarrow 0$ against $t \rightarrow \infty$. For the free Gaussian, the expected number of sign reversals of the two-sheet realization over all time is $\pi\varphi(1)$ for every $h > 0$ and zero at $h = 0$ (recording note, (150)); the total phase $\arctan u$ of (143) there reaches $\pi/2$ for every $h > 0$ and is zero at $h = 0$. Both are limits taken at infinite time first; at fixed finite time both vanish continuously as $h \rightarrow 0$. The first is a hidden count and the second a phase, so neither is a record; their observable counterpart is the breakdown of the classical limit after the Ehrenfest time, which Theorem 3 exhibits at the fold time t_* for inertial motion. The all-time limit therefore supports the thesis only through folding, which is already Theorem 3.

6. What the thesis is, and what it needs

- **Galileo's comparison with statistical preparations has no gap.** Theorem 1 makes the record family constant in h . Any gap argument run on the parabola with Gaussian preparations and marks will fail, for the same reason the deformation families fail.
- **Determinate preparations that fold have the gap, in complete records.** Theorems 2 and 3 prove it on every finite partition and exactly for inertia with graded velocity. Newton's general forces fold localized determinate preparations in finite time, since the period depends on the amplitude; central forces (Book I, Proposition I) are the natural next case. Beyond Galileo the thesis holds in this topology.

- **Smeared records and statistical preparations reach zero.** This is the content of the weak-limit theorems named in §1, and of (4).
- **Refinement with records reverses the roles.** The zero theory is the regular side of that limit.
- **The thesis is an order of limits.** By the Corollary to Theorem 2, a record of resolution δ reaches Newton iff $h/(\delta\Delta p) \rightarrow 0$. “Complete records” means $\delta \rightarrow 0$ before $h \rightarrow 0$: the record limit is taken first, as the continuum limit is taken first in the mass gap.

The thesis is therefore exactly as strong as two premises: that records are complete, the densities themselves rather than their smearings, and that preparations are determinate, one velocity at each place. The second is the fifth postulate’s joint determinacy (fifth-postulate note) read as a preparation, Newton’s *velocitas ultima* assigned to every particle of the ensemble. The first is the records reading already required by Leibniz continuity. The gap claim and the Leibniz claim rest on the same reading of records, and the gap claim adds determinacy of the preparation. Under those two premises the statement “Newton exists only with a positive scale” has the precise form: the construction without branch data exists for every $h > 0$ and refines without extra data (1998-conjecture note, Theorem 2(b)), while at $h = 0$ it needs a branch rule (Theorem 1 there) and, after a fold, is not the limit of the $h > 0$ records (Theorems 2–3 here). With either premise dropped, the zero branch is reached.

7. Consequence for STATE

The gap thesis is located: it holds for complete records of determinate preparations beyond their first fold, and fails for Galileo’s comparison with Gaussian preparations and marks, for smeared records, and for refinement with records. Next gap lemma: the first fold of a determinate preparation under a central force on Newton’s polygon (Book I, Proposition I), its fold time, and the non-convergence of its complete record by Theorem 2. The two premises, complete records and determinate preparation, become the stated hypotheses of the gap, to be defended with the Leibniz records reading. Reject a gap argument run inside a deformation family, on Gaussian preparations of the parabola, or on smeared records, and reject presenting the non-commutation of $h \rightarrow 0$ with $t \rightarrow \infty$ as a record without an observable.